

Indium, germanium, gallium and other trace elements in sphalerite, enargite and colusite from high-sulfidation polymetallic mineralization: preliminary data from the Colquijirca district, central Peru

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Abstract. Porphyry-related (Cordilleran-type) polymetallic epithermal deposits are a major source of base and precious metals, and may host significant contents of critical elements such as In, Ge and Ga. Among the analyzed minerals (sphalerite, enargite and colusite) from high-sulfidation epithermal main ore stage mineralization at Colquijirca, laser ablation inductively coupled plasma mass spectrometry (LA-ICP-MS) measurements indicate that In and Ge are mainly concentrated in sphalerite (interquartile range [IQR] 2775-899 ppm, up to 3643 ppm In; IQR = 63-26 ppm, up to 69 ppm Ge). Gallium contents are low in the three analyzed minerals (highest Ga content in colusite: IQR = 17-0.14 ppm, up to 17 ppm). Smooth ablation profiles and molar content correlations suggest that In, Ge and Ga were incorporated via coupled substitution involving other elements found in trace amounts in the crystal lattice. These new and available data on other deposits support that high-sulfidation epithermal mineralization may represent a potential source of critical metals and metalloids.

1 Introduction

The Colquijirca mining district, central Peru, contains one of the largest Cordilleran-type polymetallic [Cu-Zn-Pb-(Au-Ag)] resources worldwide (199.6 Mt Cu-Au-Ag ore and 168.6 Mt Zn-Pb-Ag ore, Casanova 2018). In 2020, Sociedad Minera El Brocal, subsidiary of Compañía de Minas Buenaventura, registered a production of 30,608 metric tons Cu, 54,909 t Zn and 20,066 t Pb (Compañía de Minas Buenaventura 2020).

Previous studies have shown that porphyry-related Cordilleran-type polymetallic mineralization may host attractive contents of the so-called critical metals and metalloids including In, Ge and Ga (see European Commission 2020 for metal criticality). Surveys specifically focused on the contents and distribution of trace elements in ore minerals from polymetallic epithermal deposits in Peru have been recently performed for the Morococha district (Benites et al. 2021a) and the Ayawilca deposit (Benites et al. 2021b).

These studies reveal that several ore minerals in particular sphalerite, stannite and chalcopyrite in low-, intermediate-, and high-sulfidation assemblages of Cordilleran-type deposits may represent an attractive source for these elements.

In this work, a preliminary study of trace element contents in ore minerals from a predominantly high-sulfidation assemblage in the Smelter deposit, Colquijirca district, is presented. In situ major and trace-element content determinations in enargite, colusite and sphalerite were performed by electron microprobe (EPMA; University of Barcelona) and laser ablation-inductively coupled plasma-sector field-mass spectrometry (LA-ICP-SF-MS; ETH Zürich) using the same equipment and analytical procedure as in Benites et al. (2021a).

2 Geological setting

The Colquijirca district is part of the Oligocene-Miocene porphyry-related metallogenic belt of central and northern Peru (Fig. 1). At Colquijirca the deposits are emplaced in and adjacent to a middle Miocene dacitic diatreme-dome (12.43 ± 0.06 Ma to 12.12 ± 0.03 Ma, $^{40}\text{Ar}/^{39}\text{Ar}$ on biotite, BendeZú et al. 2003; U-Pb on zircon, Baumgartner et al. 2009) intruded in Paleozoic basement and Mesozoic to Cenozoic sedimentary rocks (BendeZú and Fontboté, 2009). Cordilleran-type polymetallic mineralization is the economically most important in the district and includes the Smelter and Colquijirca deposits both located north of the diatreme-dome complex and the San Gregorio deposit south of it constituting an at least 7.5-km-long N-S mineralized corridor. The sample studied in the present contribution comes from the most internal zone of the "Main Ore Stage" recognized in the Smelter and Colquijirca deposits (BendeZú and Fontboté 2009) and dated at 10.83 ± 0.06 to 10.56 ± 0.06 Ma ($^{40}\text{Ar}/^{39}\text{Ar}$ on alunite; BendeZú et al. 2008). This internal zone consists mainly of enargite ($\pm$ luzonite, pyrite, colusite, tennantite, quartz). It grades into a zone dominated by bornite, chalcopyrite and tennantite that in turn grades into a Pb-Zn sulfide zone surrounded by Fe-Mn-Zn carbonates and

iron oxides. The polymetallic mineralization is interpreted to result from low to intermediate-density single-phase fluids of magmatic origin diluted in places by meteoric waters (Bendezú 2007; Bendezú and Fontboté 2009; Casanova 2018).



Figure 1. Location of the Colquijirca district. Modified from Scherrenberg et al. (2016) and Fontboté (2018). Abbreviations: EC = Eastern Cordillera; MFTB = Marañón Fold-Thrust Belt; WC = Western Cordillera.

3 Results

3.1 Ore mineralogy and textures

The studied ore sample from the Smelter deposit belongs to the enargite zone of the main ore stage (Bendezú and Fontboté 2009). The hypogene assemblage is dominated by coarse prismatic subhedral aggregates of enargite. Trace amounts of colusite, luzonite, an undetermined CuAsSb sulfosalt, pyrite, marcasite, sphalerite, arsenopyrite, tennantite and chalcopyrite are observed. Fe-poor sphalerite occupy interstitial space between enargite crystals (Fig. 2A), and late arsenopyrite occurs in open spaces. Colusite and luzonite occur intergrown with enargite grains (Fig. 2B). Pyrite-marcasite are observed as disseminations in enargite and as inclusions in sphalerite. Tennantite occurs as disseminations in enargite and as fine veinlets along with chalcopyrite across enargite. Additionally, minute inclusions of an undetermined CuAsSb sulfosalt of greenish-gray color and strong anisotropy are observed in enargite grains.

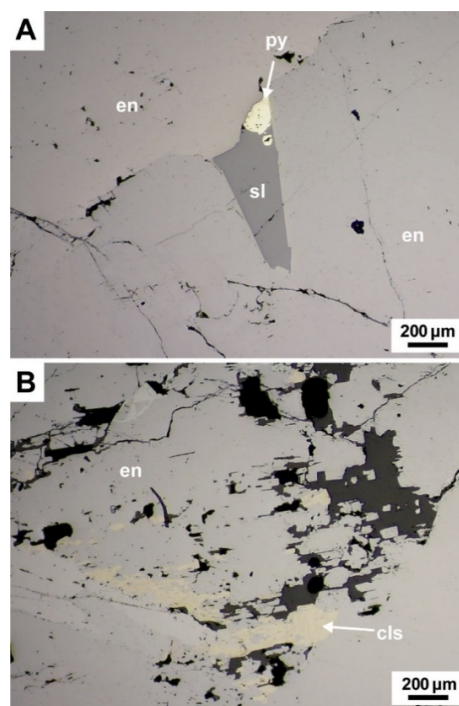


Figure 2. Photomicrographs (reflected light) of studied sample from a high-sulfidation assemblage in the Smelter deposit, Colquijirca district. A) Prismatic subhedral enargite (en) crystals with interstitial Fe-poor sphalerite (sl) and pyrite (py). B) Enargite grain intergrown with colusite (cls).

3.2 Ore mineral geochemistry

Sphalerite yields very low Fe contents (interquartile range [IQR] = 18-16 ppm, up to 18 ppm), values consistent for an enargite-bearing high sulfidation assemblage, while Cu (IQR = 8330-5203 ppm) and Cd (IQR = 5413-1728 ppm) are much higher. Regarding critical elements, sphalerite shows relatively high In contents (IQR = 2775-899 ppm, up to 3643 ppm) and low Ge (IQR = 63-26 ppm, up to 69 ppm) and Ga (IQR = 13-4.0, up to 14 ppm). Smooth LA-ICP-MS profiles (Fig. 3A) and a positive correlation between Cu and In at 1:1 molar proportions suggests a $\text{Cu}^+ + \text{In}^{3+} \leftrightarrow 2\text{Zn}^{2+}$ coupled substitution (cf. Cook et al. 2009).

In enargite, Sb values (IQR = 1.7-1.2 wt%) are higher than those of Sn (IQR = 8445-2716 ppm, up to 1.1 wt%), Fe (IQR = 311-84 ppm, up to 1088 ppm), Hg (IQR = 261-78 ppm, up to 326 ppm), and Zn (IQR = 68-29 ppm, up to 125 ppm). Germanium (IQR = 9.7-3.2 ppm, up to 23 ppm) and Ga (IQR = 0.18-0.14 ppm, up to 0.61 ppm) values are low, and In is systematically below detection limit. Smooth ablation profiles (Fig. 3B) and the positive correlation between Hg, Fe and Zn with Sn (Fig. 4) suggest a $\text{Sn}^{4+} + [\text{Hg,Fe,Zn}]^{2+} \leftrightarrow \text{As}^{5+} + \text{Cu}^+$ coupled substitution as described by Liu et al. (2019). In contrast, irregular ablation profiles for Au (up to 22.7 ppm), Ag (up to 205 ppm), Pb (up to 11.5 ppm) and Bi (up to 345 ppm; Fig. 3B) suggest the presence of micro inclusions of discrete minerals (e.g. Au, galena), as also noticed by Deyell and Hedenquist (2011) and Liu et al. (2019) in enargite from the Mankayan district, Philippines, and the Zijinshan orefield, South China, respectively.

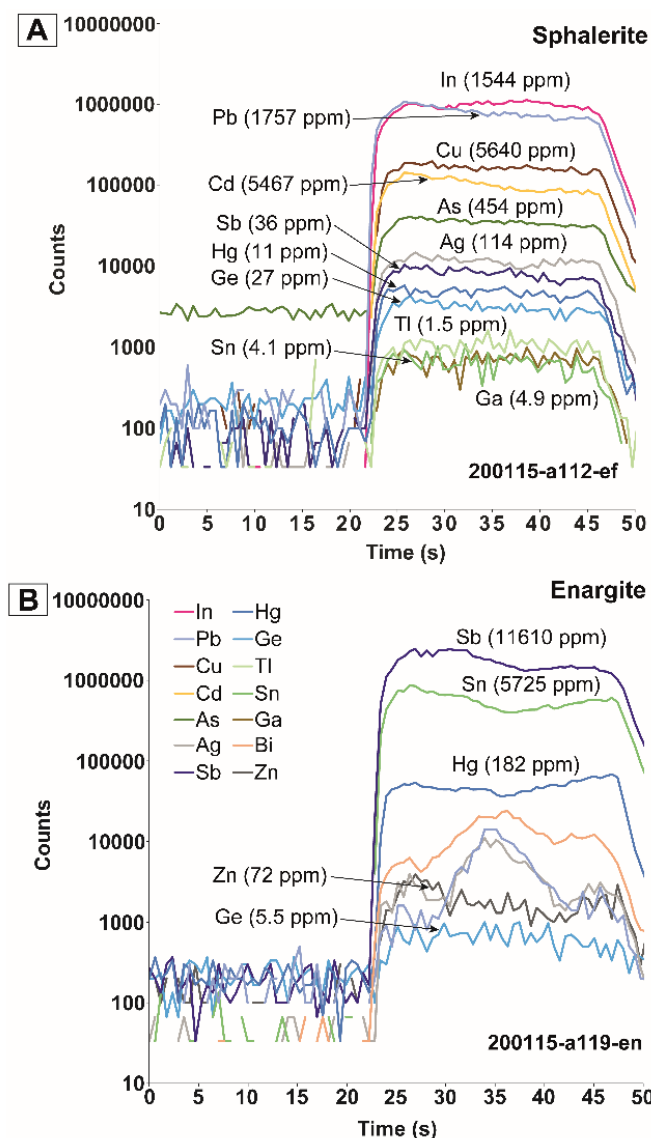


Figure 3. Time-resolved LA-ICP-MS profiles for selected spots showing only trace and minor elements on A) sphalerite (200115-a112-ef) and B) enargite (200115-a119-en) from the Smelter deposit, Colquijirca district.

In colusite, As (IQR = 8.9-7.1 wt%, up to 8.9 wt%) and Sn (IQR = 8.5-7.25 wt%, up to 8.49 wt%) contents are higher than Sb contents (IQR = 2.2-1.4 wt%, up to 2.2 wt%). The average empirical formula is $\text{Cu}_{12.73}(\text{As}_{1.76}\text{Sn}_{1.05}\text{Sb}_{0.06})_{\Sigma=2.87}\text{S}_{16}$. Other elements at relatively high values in analyzed colusite grains include Fe (IQR = 8825-739 ppm, up to 9288 ppm), Zn (IQR = 254-92 ppm, up to 326 ppm), Se (IQR = 167-60 ppm, up to 185 ppm) and Hg (IQR = 593-82 ppm, up to 898 ppm). Colusite shows moderate to low Ge (IQR = 56-5.7 ppm, up to 59 ppm) and Ga (IQR = 17-0.14 ppm, up to 17 ppm) contents. The content of In could not be evaluated due to isobaric interference with Sn, which is a major element in this mineral.

4 Discussion and conclusions

Sphalerite from the studied high-sulfidation mineral assemblage in the Smelter deposit, Colquijirca district, yielded In contents (IQR = 2775-899 ppm, up to 3643 ppm) slightly higher than those reported for sphalerite from epithermal veins dominated by high-sulfidation assemblages overprinting the Toromocho porphyry stock in the Morococha district (IQR = 1291-726 ppm, up to 1456 ppm; Benites et al. 2021a). At the Morococha deposit, the highest In contents were found, indeed, in sphalerite from the aforementioned high-sulfidation assemblage relatively to sphalerite from low- (e.g. Manto Italia, IQR = 324-261 ppm, up to 459 ppm) and intermediate-sulfidation (e.g. Morro Solar, IQR = 471-8.0 ppm, up to 1804 ppm) assemblages. At the Cerro de Pasco deposit, in contrast, sphalerite from high- to intermediate-sulfidation assemblages yields lower In content (IQR = 20-0.89 ppm, up to 104 ppm) than sphalerite from low-sulfidation assemblages (255-29 ppm, up to 1449 ppm; Rottier et al. 2018). In the Mt Carlton epithermal deposit in Australia, sphalerite from an enargite-bearing high-sulfidation assemblage (up to 571 ppm In; Sahlström et al. 2017) also yielded lower In contents than sphalerite from an intermediate-sulfidation assemblage (up to 2169 ppm In). Much higher In contents have been described in Fe-rich sphalerite in low-sulfidation assemblages at the Ayawilca deposit (up to 1.7 wt%; Benites et al. 2021b) and in intermediate-sulfidation assemblages in several Bolivian polymetallic deposits (e.g. up to 10 wt% in the Ánimas-Chocaya-Siete Suyos district; Torró et al. 2019). Therefore, it appears that the sulfidation state of the mineralizing fluid does not exert a dominant control on In enrichment in sphalerite. Benites et al. (2021a) proposed that metal-rich magmatic-hydrothermal fluids with a high availability of Cu favor the incorporation of In in sphalerite through the coupled substitution described above; dilution of the magmatic fluids and crystallization of Cu-bearing phases would reduce the capacity for In incorporation in sphalerite whatever the sulfidation state.

Enargite grains from the Smelter deposit yield relatively low Ge contents (IQR = 9.7-3.2 ppm, up to 23 ppm) if compared with previous reports on other epithermal deposits worldwide (e.g. up to 3461 ppm in Zijinshan, China, Liu et al. 2019; up to 2189 ppm in Mt Carlton, Australia, Sahlström et al. 2017). Indium content in analyzed enargite could not be determined, but it is generally low in enargite from other high-sulfidation epithermal mineralization (e.g. up to 30 ppm in Zijinshan, China, Liu et al. 2019; up to 5.8 ppm in Cerro de Pasco, Peru, Rottier et al. 2018). According to Deyell and Hedenquist (2011), the concentration of trace elements in enargite reflects the composition and temperature of mineralizing fluids, which may be influenced by the distance from the magmatic-hydrothermal center, the fluid ascent path, and the degree of wall-rock interaction.

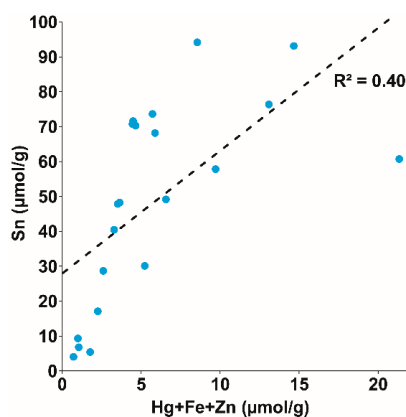


Figure 4. Binary plot illustrating the correlation between Hg+Fe+Zn and Sn in enargite from the Smelter deposit, Colquijirca district.

Analyzed colusite from the Smelter deposit yields low Ga and Ge values. However, high-sulfidation deposits from the Cretaceous Panagyurishte district of Bulgaria locally contain germanocolusite $[\text{Cu}_{13}\text{V}(\text{Ge},\text{As})_3\text{S}_{16}]$ (Terziev 1966). We are not aware of reports on In and Ga contents in colusite from other polymetallic epithermal deposits. In volcanogenic massive sulfide deposits, high Ge content in colusite has been reported at the Waterloo deposit in Australia (up to 5.0 wt%; Wagner and Monecke 2005), the Yanahara mine in Japan (up to 4.3 wt%; Kase et al. 1994), and the Ezuri Kuroko deposit in Japan (up to 4.1 wt%; Komuro and Kajiwarra 2004).

This preliminary study provides new data on trace element composition of sphalerite, colusite and enargite in high-sulfidation assemblages in Cordilleran-type polymetallic mineralization. High In contents in analyzed sphalerite adds to previous reports that indicate that sphalerite from high-sulfidation assemblages might be attractive for this critical metal. A study with a larger set of representative samples along the Colquijirca district could facilitate comparison with the study on the distribution of In, Ge, Ga and other minor and trace elements on the also Cordilleran-type polymetallic deposits of the Morococha district performed by Benites et al. (2021a).

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